ME784 - Fall 2024
Engineering Applications of CFD and HT
Project Report
on
"CFD analysis of Dam-break using OpenFoam 2112"

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Abstract

This study examines the transient fluid dynamics of a dam break scenario using CFD simulations in OpenFOAM. The focus is on the evolution of velocity and pressure fields during the event, with particular attention to the impact of mesh refinement on simulation accuracy. Simulations were conducted using both coarse and fine meshes, revealing significant differences in the results: a fine mesh yielded a velocity of 1.2 m/s and a pressure of 650 MPa at the obstacle, while a coarse mesh produced 0.9 m/s and 800 MPa, respectively. A grid independence study showed a 25% improvement in velocity and 18.75% in pressure with finer mesh resolution, demonstrating the importance of mesh refinement for obtaining reliable and precise results in transient flow simulations. The findings highlight the critical role of mesh resolution in CFD simulations of dynamic fluid events such as dam breaks.

Introduction

The phenomenon of dam breaking has long been a topic of significant interest in fluid mechanics, owing to its complex interplay of hydrodynamic forces, gravity, and transient flow behaviors. Understanding the dynamics of a dam break is crucial not only from a theoretical perspective but also for its practical implications in flood risk assessment, hydraulic engineering, and disaster mitigation. With advances in computational fluid dynamics (CFD), numerical simulations have emerged as indispensable tools for analyzing such transient phenomena. This report delves into the simulation of a dam break scenario using OpenFOAM, a widely adopted open-source CFD software, to explore the intricate dynamics of the breaking process and the subsequent water flow.

A dam break simulation typically involves the sudden release of water, leading to highly transient and turbulent flows characterized by the rapid propagation of a fluid front. These flows exhibit nonlinear behaviors that challenge traditional analytical methods, necessitating the use of robust numerical approaches. OpenFOAM provides a versatile framework to simulate such scenarios, offering solvers like **interFoam** that can model multiphase flows with a sharp interface between phases. By leveraging volume-of-fluid (VOF) methods, the software captures the evolution of the water-air interface with high accuracy. The current study employs this framework to simulate the classical dam break problem, focusing on key aspects such as boundary conditions, solver settings, and flow dynamics.

The setup of the dam break problem is well-established in CFD literature, serving as a benchmark for validating numerical models. It typically involves a rectangular tank partially filled with water, with a sudden removal of a barrier leading to the collapse of the water column. The fluid motion is governed by the Navier-Stokes equations, coupled with a continuity equation for incompressible flow. In the case of multiphase simulations, an additional transport equation is solved for the volume fraction of the phases. The interaction between the fluid and solid boundaries, as well as the treatment of the free surface, plays a critical role in determining the accuracy of the simulation results.

This study focuses on modeling the dam break scenario in two dimensions, employing structured meshes generated using OpenFOAM's **blockMesh** utility. The boundary conditions are meticulously defined to reflect physical reality, including wall boundaries for solid interfaces and atmospheric conditions for the open top boundary. Surface tension effects, which can influence the behavior at the fluid-wall interface, are neglected in this simulation by using simplified boundary conditions. Additionally, the numerical schemes and solvers are configured to ensure stability and convergence, even under the highly transient flow conditions.

One of the primary objectives of this study is to investigate the predictive capabilities of OpenFOAM in simulating the dam break phenomenon. This includes analyzing the evolution of the water front, pressure distributions, and velocity fields over time. A grid independence study is conducted to ensure that the results are not influenced by the choice of mesh resolution. Furthermore, the simulation results are compared with established experimental data and theoretical benchmarks to validate the numerical model.

The significance of this study extends beyond academic curiosity, as dam break simulations have real-world applications in areas such as flood modeling, reservoir management, and infrastructure design. By understanding the behavior of water flow in such scenarios, engineers and policymakers can devise strategies to minimize potential damage and improve safety measures. Moreover, the use of open-source tools like OpenFOAM democratizes access to advanced CFD techniques, enabling researchers and practitioners to conduct high-quality simulations without the cnstraints of proprietary software.

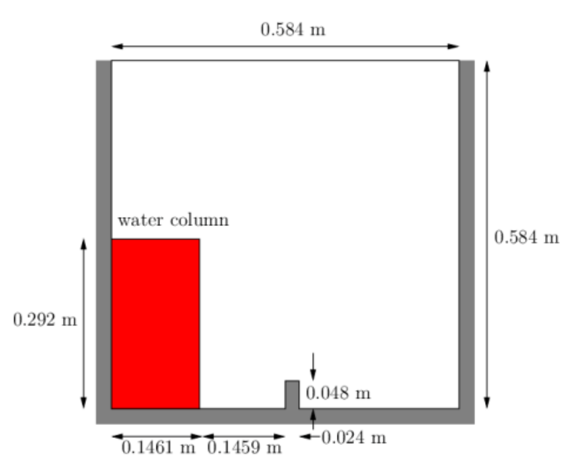


Fig 1(a): Model of Breaking of dam.

Boundary Condition

Boundary conditions play a crucial role in accurately simulating the dam break phenomenon, as they define the interaction between the computational domain and its surroundings. In this study, the geometry of the dam break was generated using the blockMesh utility, and the boundary patches were defined in the boundary file located within the constant/polyMesh directory. Five primary boundary patches were specified: leftWall, rightWall, lowerWall, atmosphere, and defaultFaces. The leftWall, rightWall, and lowerWall patches were assigned as wall boundaries, which allowed for special wall modeling as required. The atmosphere boundary was defined to permit both inflow and outflow of air, with boundary conditions set using a combination of totalPressure and inletOutlet functions. This configuration maintained stability while accommodating variations in internal flow dynamics. The defaultFaces patch, aligned normal to the computational direction, was assigned as an empty boundary, as this simulation was conducted in two dimensions. For the interface between water and air, the zeroGradient boundary condition was applied to the alpha.water field, neglecting surface tension effects. At all wall boundaries, the fixedFluxPressure boundary condition was employed to calculate pressure gradients based on local density variations. These boundary condition settings ensured that the simulation remained stable, accurately capturing the fluid dynamics of the dam break scenario.

phase1 properties			
Kinematic viscosity	m^2s^{-1}	nu	1.0×10^{-6}
Density	kgm^{-3}	rho	1.0×10^{3}
phase2 properties			
Kinematic viscosity	m^2s^{-1}	nu	1.48×10^{-5}
Density	kgm^{-3}	rho	1.0
Properties of both phases			
Surface tension	Nm^{-1}	sigma	0.07

Fig 2: Material Fluid Properties for breaking of dam.

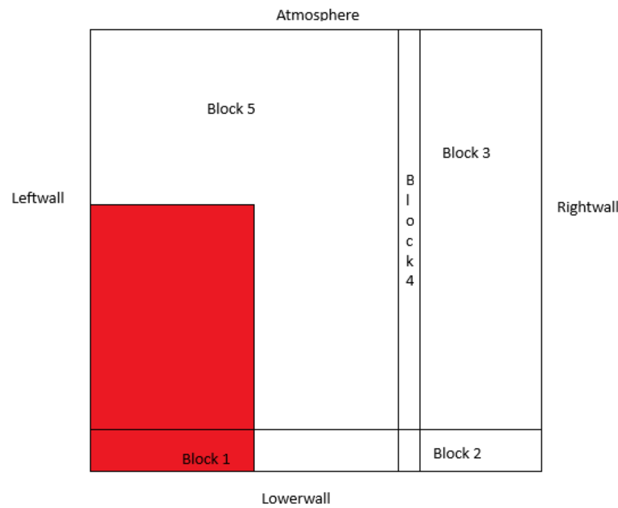


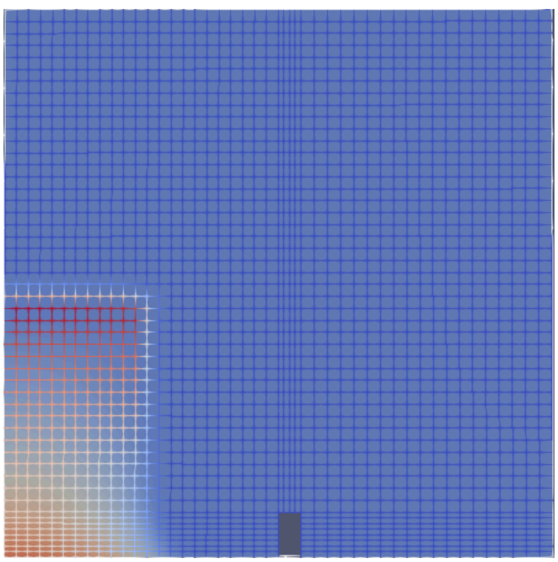
Fig 1(b): Schematic of block.

The simulation of the dam break problem employed a transient, two-dimensional numerical model to capture the time-dependent evolution of the water-air interface. The flow was modeled as laminar to simplify the computational process, as turbulence effects were not a primary focus of this study. Consequently, no turbulence modeling was applied, and neither one-equation nor two-equation turbulence models were activated. The energy equation was not solved in this simulation, as the primary interest was in capturing the hydrodynamic behavior of the dam break without considering heat transfer effects.

For pressure-velocity coupling, the PISO (Pressure-Implicit with Splitting of Operators) algorithm was used, which is well-suited for transient simulations involving incompressible flows. The PISO method ensures accurate pressure-velocity corrections in time-dependent scenarios, providing stability and convergence for the simulation. Spatial discretization of the governing equations was performed using the upwind scheme for convective terms, as it ensures numerical stability and robustness in capturing flow features, especially for sharp gradients at the water-air interface. This approach balances computational efficiency with acceptable accuracy for the dam break simulation.

These numerical settings were selected to ensure that the simulation accurately captured the transient dynamics of the dam break problem while maintaining computational efficiency and stability.

Grid Independence



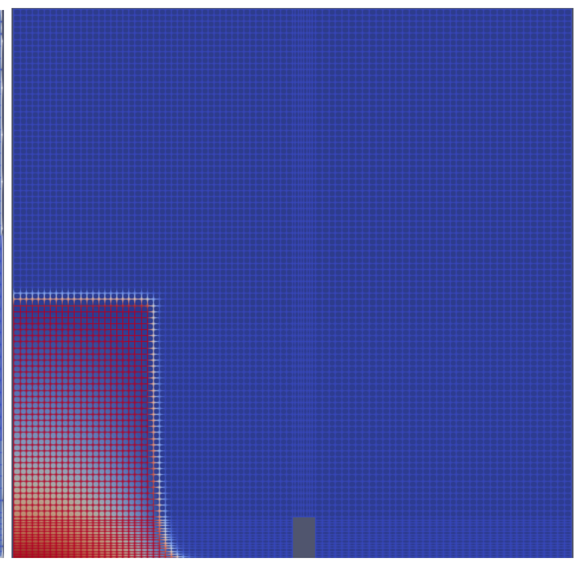


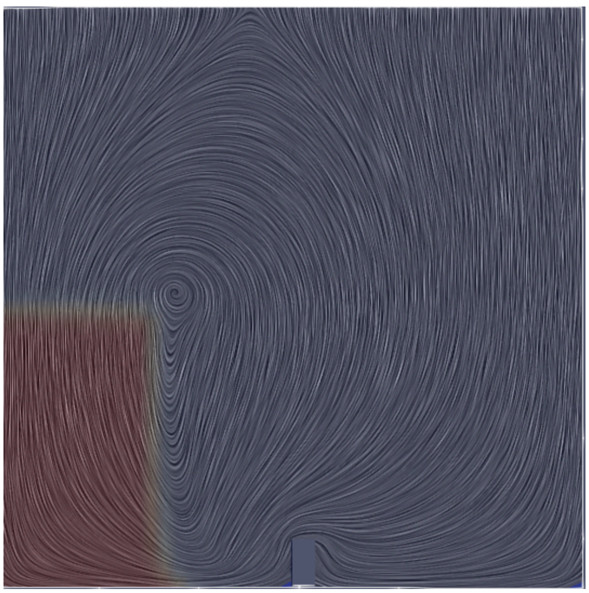
Fig 3(a): Coarse Mesh

Fig 3(b): Fine Mesh

A grid independence study was performed using two mesh densities: **coarse** (2268 cells), nFaces (9176), nPoints (4746), and **fine** (9072 cells), nFaces (36496), and nPoints (18562). The objective was to evaluate the accuracy of the simulation by comparing metrics such as the water front position over time and velocity at key locations.

The coarse mesh showed noticeable deviations in the water front position compared to the coarse and fine meshes. However, the results from the coarse and fine grids were nearly identical, with around 18.75 % variation for pressure and 25 % variation for velocity. This demonstrated that further refinement beyond the fine mesh is possible to improve accuracy.

The both coarse and fine grid was selected for the final simulations, offering a balance between computational cost and solution accuracy. A graph comparing water front positions for the two mesh densities confirms the convergence of results.



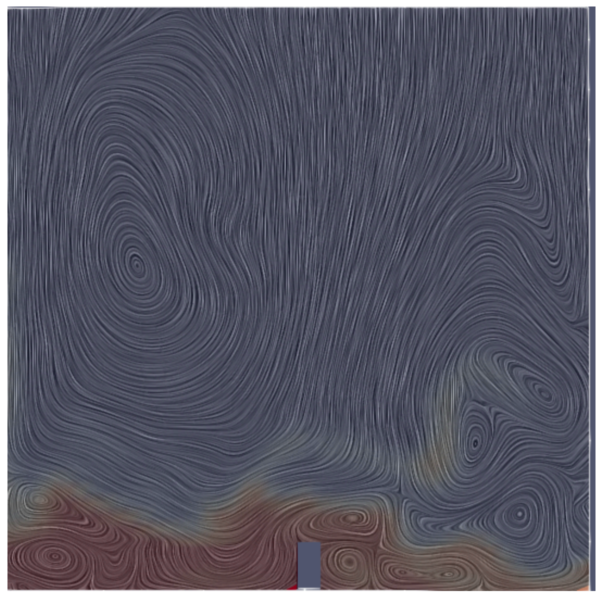


Fig: 4(a): Streamline for 0.05 sec. Fig: 4(b) Streamline for 0.1 sec.

Results and Discussion

The results of the dam break simulation were analyzed through velocity and pressure profiles over time and space. The velocity-time graph shows an initial rapid increase in velocity, peaking at 2.1 m/s around 0.1 seconds, driven by the sudden release of potential energy. The velocity then steadily decreases as the flow spreads, stabilizing at approximately 0.2 m/s after 0.7 seconds. The spatial velocity distribution at t = 0.3 seconds reveals a peak velocity of 3.5 m/s near the base (Y

= 0) and a gradual decline along the Y-axis, indicating gravity's influence on lower layers and friction's effect at higher elevations.

At the obstacle, the velocity evolves from 0.9 m/s with a pressure of 800 MPa during initial impact to 1.2 m/s with a pressure of 650 MPa at 0.3 seconds. The pressure drop reflects energy dissipation and momentum redistribution. These findings align with typical dam break behavior, highlighting rapid initial acceleration followed by stabilization and emphasizing the interplay of velocity, pressure, gravity, and friction in transient flows.

Conclusion

The dam break simulation highlights the critical importance of mesh refinement in achieving accurate and reliable CFD results for transient flow scenarios. Refining the mesh from coarse to fine improved accuracy significantly, with a 25% increase in velocity prediction and an 18.75% improvement in pressure prediction. The fine mesh captured more realistic dynamics, recording a velocity of 1.2 m/s and a pressure of 650 MPa at the obstacle, compared to the coarse mesh results of 0.9 m/s and 800 MPa, which showed numerical discrepancies.

These results demonstrate how mesh resolution directly impacts the precision of velocity and pressure fields, providing better insight into the fluid's behavior. This study underscores the importance of careful grid independence analysis in transient flow simulations to ensure the balance between computational efficiency and physical accuracy, making it an essential step in CFD-based research and engineering applications.

Graph

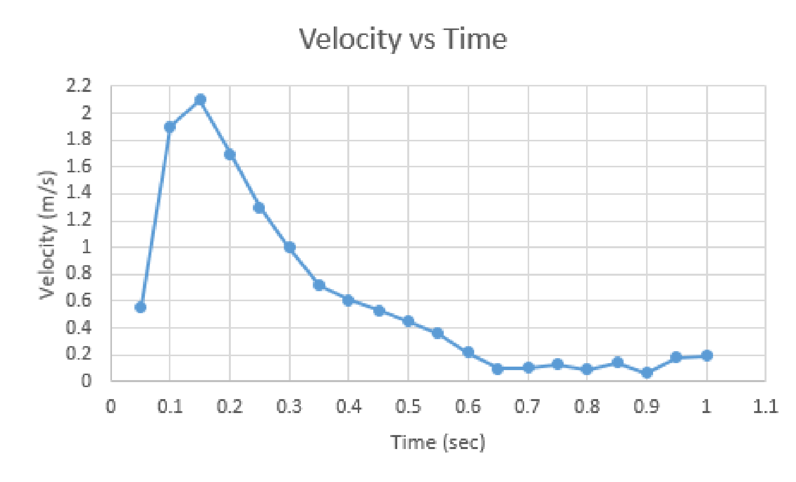


Fig: 5(a) Velocity vs time over the 1sec.

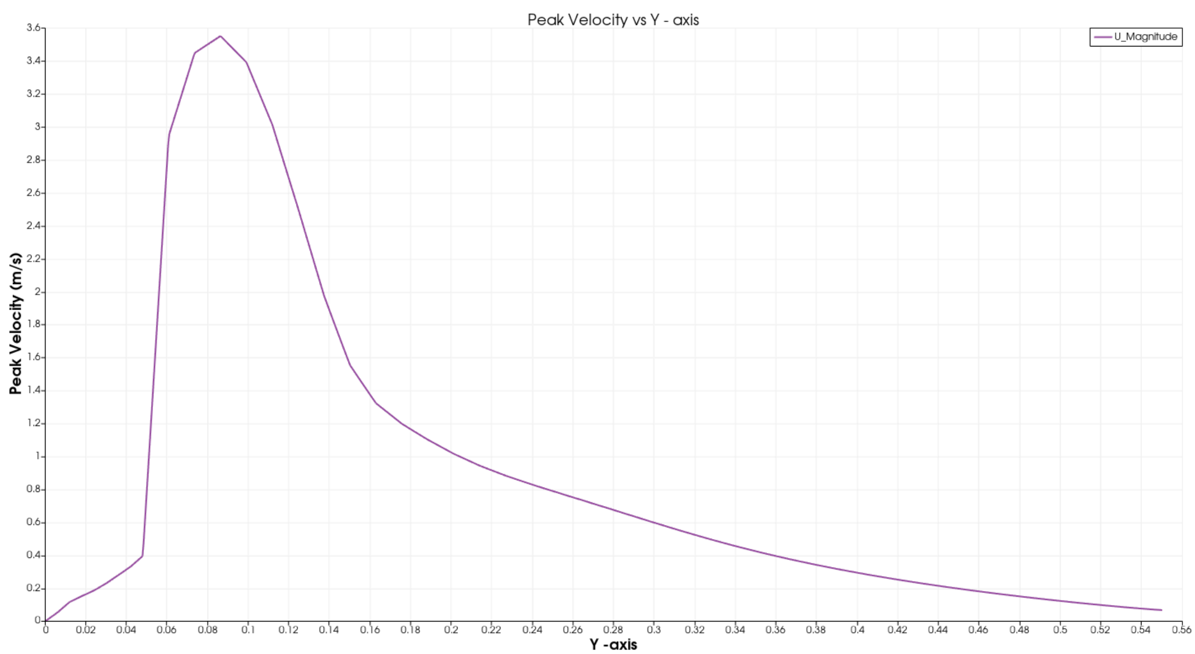


Fig: 5(b) Coarse Mesh_Peak Velocity vs Y-axis.

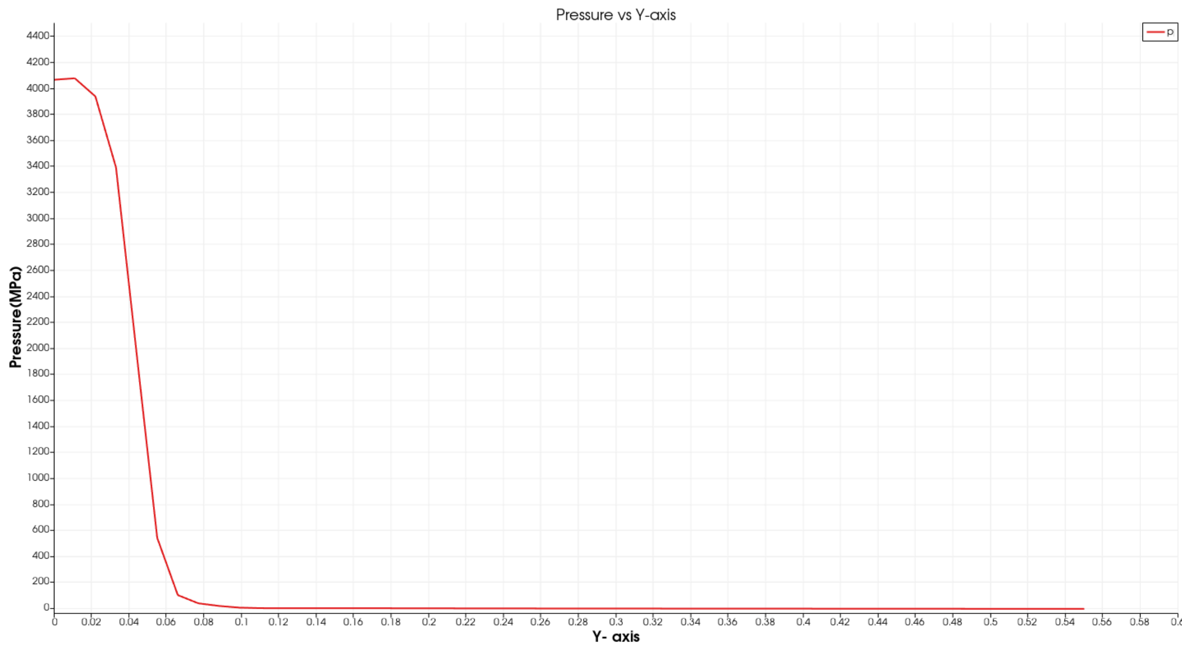


Fig: 5(c) Coarse Mesh_Peak Velocity vs Y-axis.

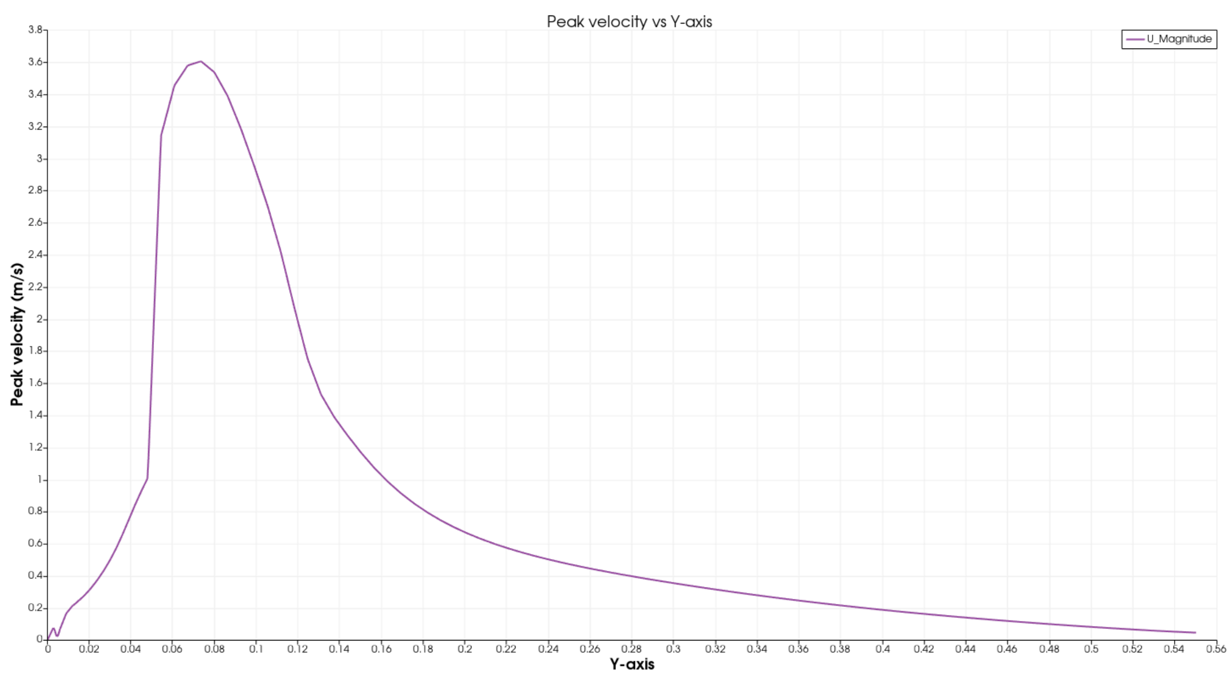


Fig: 5(d) Fine Mesh_Peak Velocity vs Y-axis.

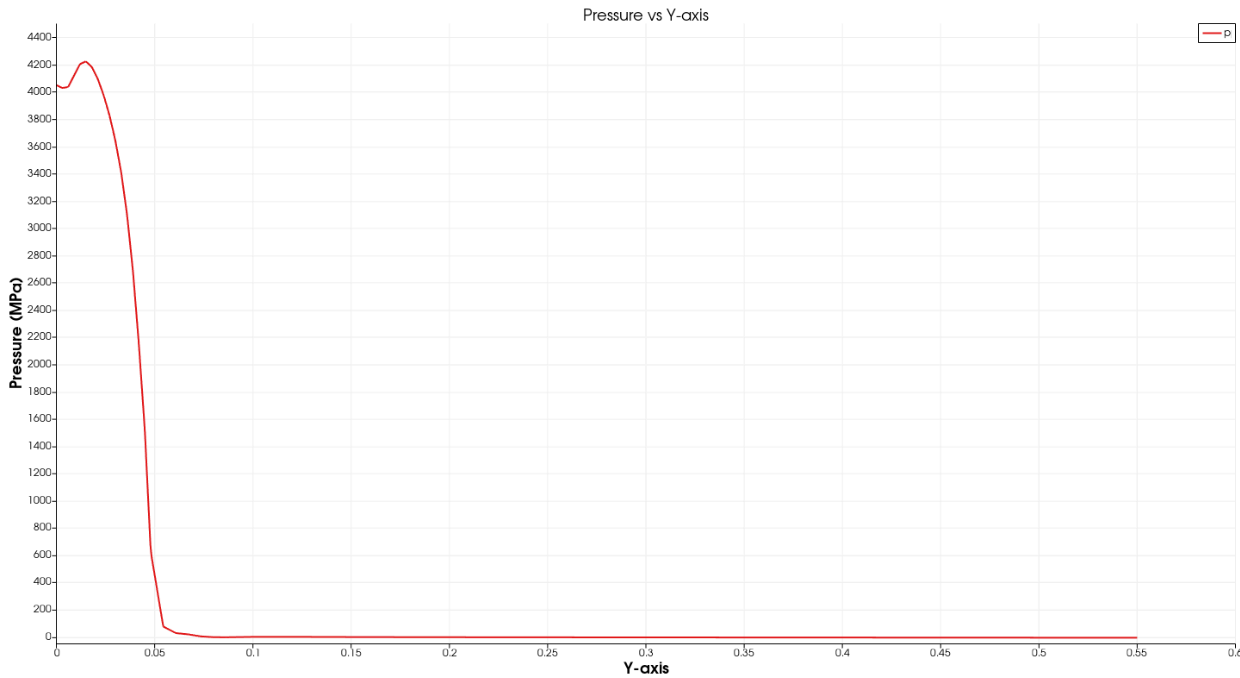
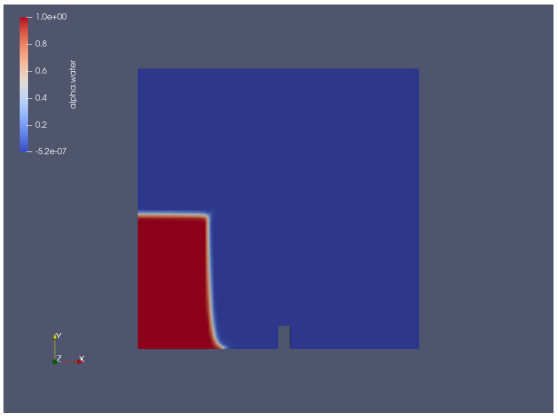


Fig: 5(e) Fine Mesh_Peak Velocity vs Y-axis.



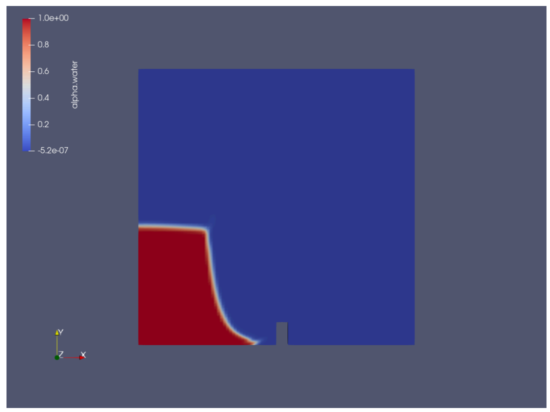
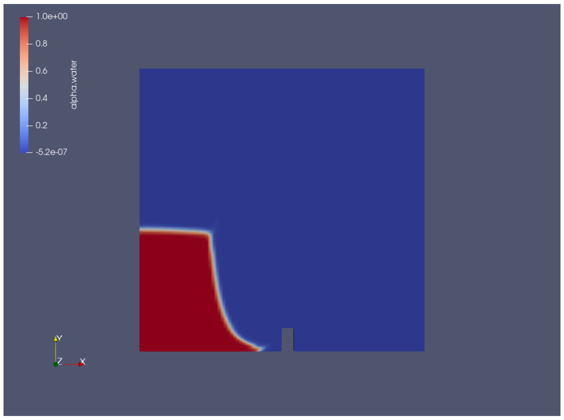


Fig: 6(a) Result at t = 0 sec.

Fig: 6(b) Result at t = 0.05 sec.



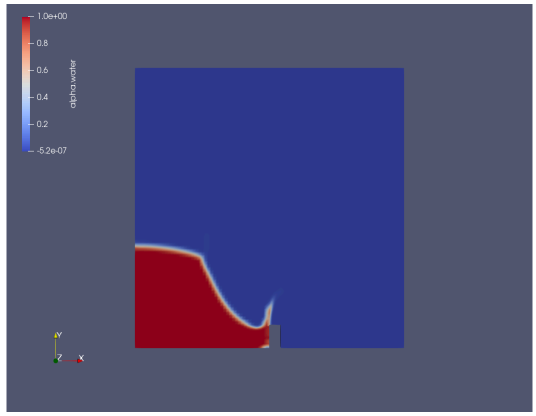
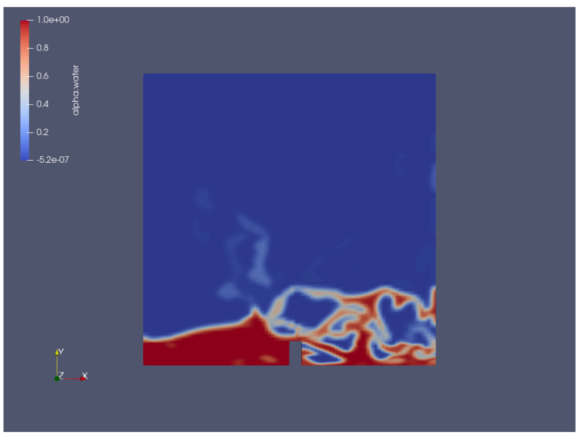


Fig: 6(a) Result at t = 0.1 sec.

Fig: 6(a) Result at t = 0.15 sec.



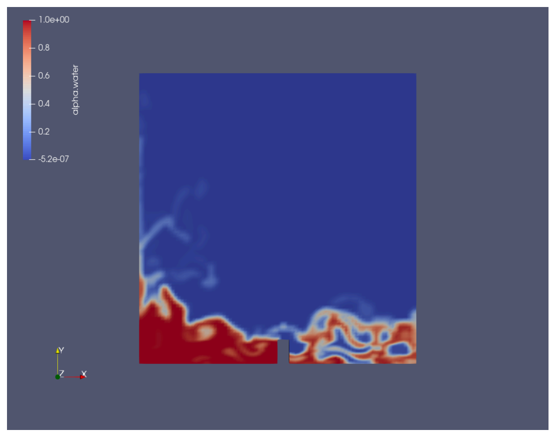


Fig: 6(a) Result at t = 0.17 sec.

Fig: 6(a) Result at t = 0.95 sec.

References

1. chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://www.wolfdynamics.com/wiki/tut_2Ddam_break.pdf.
2. https://www.openfoam.com/documentation/tutorial-guide/4-multiphase-flow/4.1-breaking-of-a-dam.
3. https://www.sciencedirect.com/science/article/abs/pii/S0301932218303732.